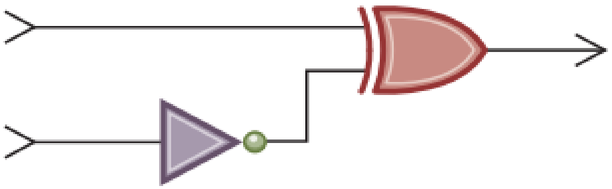
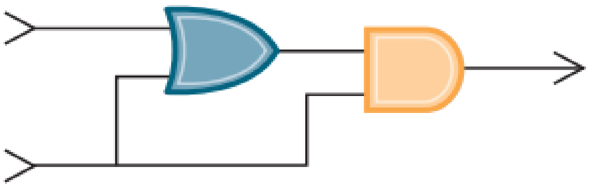
Question

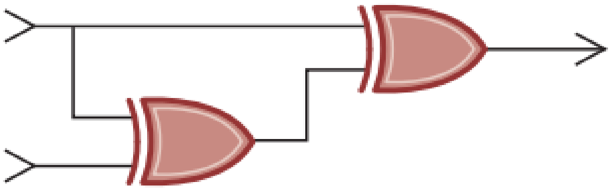
1. Determine the output of each of the following circuits, assuming that the upper input is 1 and the lower input is 0. What would be the output when the upper input is 0 and the lower input is 1?



a.



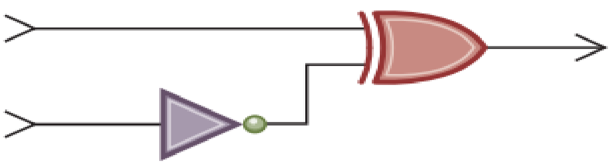
b.



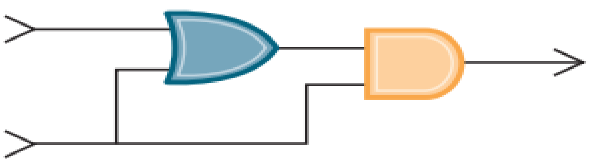
c.

Question

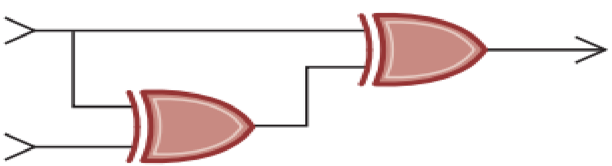
1. Determine the output of each of the following circuits, assuming that the upper input is 1 and the lower input is 0. What would be the output when the upper input is 0 and the lower input is 1?



a.



b.



c.

With a 1 on the upper input and a 0 on the lower input, all circuits will produce an output 0. If instead a 0 is on the upper input and 1 is on the lower input, circuits b and c will produce an output 1, and circuit a will still produce a 0.

Question

9. Express the following bit patterns in hexadecimal notation:
   a. 101101001011010011
   b. 000111100001
   c. 1111111011011011

Question

9. Express the following bit patterns in hexadecimal notation:
   a. 101101001011010010 11
   b. 000111100001
   c. 1111111011011011

9. a. A0A b. C7B c. 0BE

Question

10. Suppose a digital camera has a storage capacity of 500MB. How many black-and-white photographs could be stored in the camera if each consisted of 512 pixels per row and 512 pixels per column if each pixel required one bit of storage?

Question

10. Suppose a digital camera has a storage capacity of 500MB. How many black-and-white photographs could be stored in the camera if each consisted of 512 pixels per row and 512 pixels per column if each pixel required one bit of storage?

10. The image consists of 1024 x 1024 = 1,048,576 pixels and therefore 3 x 1,048,576 = 3,145,728 bytes, or about 3MB. This means that about 86 images could be stored in the 256MB camera storage system. (By comparing this to actual camera storage capacities, students can gain an appreciation for the benefits of image compression techniques. Using them, a typical 256MB storage system can hold as many as 300 images.)

Question

*30. Convert each of the following two's complement representations to its equivalent base 10 representation:

a. 010101 b. 101010 c. 110110
d. 011011 e. 111001

Question

*30. Convert each of the following two's complement representations to its equivalent base 10 representation:

a. 010101
b. 101010
c. 110110
d. 011011
e. 111001

30. a. 15 b. -12 c. 12 d. -16 e. -10

Question

*31. Convert each of the following base 10 representations to its equivalent two's complement representation in which each value is represented in 8 bits:

a. −27
b. 3
c. 21
d. 8
e. −18

Question

*31. Convert each of the following base 10 representations to its equivalent two's complement representation in which each value is represented in 8 bits:

a. −27 b. 3 c. 21
d. 8 e. −18

31. a. 0001101 b. 1110011 c. 1111111 d. 0000000 e. 0010000

Question

*33. Solve each of the following problems by translating the values into two's complement notation (using patterns of 5 bits), converting any subtraction problem to an equivalent addition problem, and performing that addition. Check your work by converting your answer to base ten notation. (Watch out for overflow.)

a. $5 + 1$
b. $5 - 1$
c. $12 - 5$
d. $8 - 7$
e. $12 + 5$
f. $5 - 11$

Question

*33. Solve each of the following problems by translating the values into two's complement notation (using patterns of 5 bits), converting any subtraction problem to an equivalent addition problem, and performing that addition. Check your work by converting your answer to base ten notation. (Watch out for overflow.)

a. 5 + 1	b. 5 − 1	c. 12 − 5
d. 8 − 7	e. 12 + 5	f. 5 − 11

33. a.

```
  5               00101
+ 1   becomes   + 00001
                  00110  which represents 6
```

b.

```
  5        00101                         00101
- 1   becomes  - 00001  which converts to  + 11111
                                          00100   which represents 4
```

c.

```
 12        01100                         01100
- 5   becomes  - 00101  which converts to  + 11011
                                          00111   which represents 7
```

d.

```
  8        01000                         01000
- 7   becomes  - 00111  which converts to  + 11001
                                          00001  which represents 1
```

e.

```
 12            01100
+ 5  becomes   + 00101
                 10001   which represents -15 (overflow)
```

f.

```
  5         00101                          00101
- 11  becomes  - 01011  which converts to  + 10101
                                           11010  which represents -6
```

Question

*34. Convert each of the following binary representations into its equivalent base ten representation:

a. 11.11
b. 100.0101
c. 0.1101
d. 1.0
e. 10.01

Question

*34. Convert each of the following binary representations into its equivalent base ten representation:

a. 11.11
b. 100.0101
c. 0.1101
d. 1.0
e. 10.01

34. a. 3 3/4 b. 4 5/16 c. 13/16 d. 1 e. 2 1/4

Question

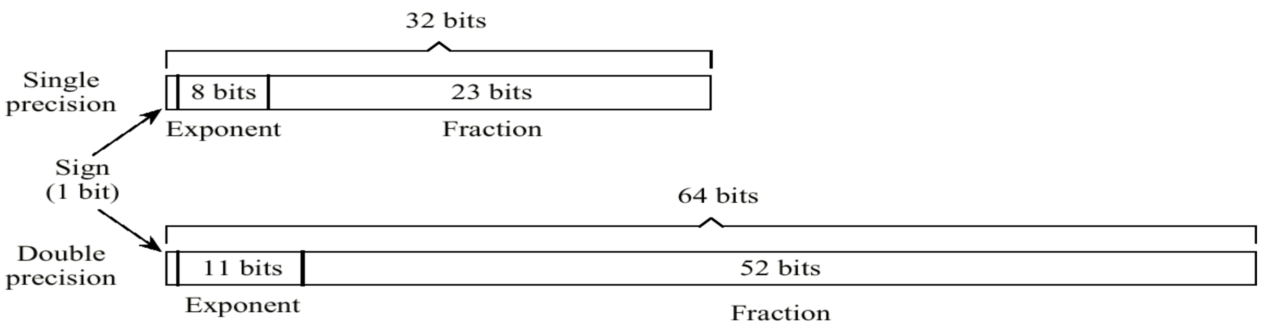
*35. Express each of the following values in binary notation:

a. $5\frac{3}{4}$ b. $15\frac{15}{16}$ c. $5\frac{3}{8}$
d. $1\frac{1}{4}$ e. $6\frac{5}{8}$

35. a. 101.11 b. 1111.1111 c. 101.011 d. 1.01 e. 110.101

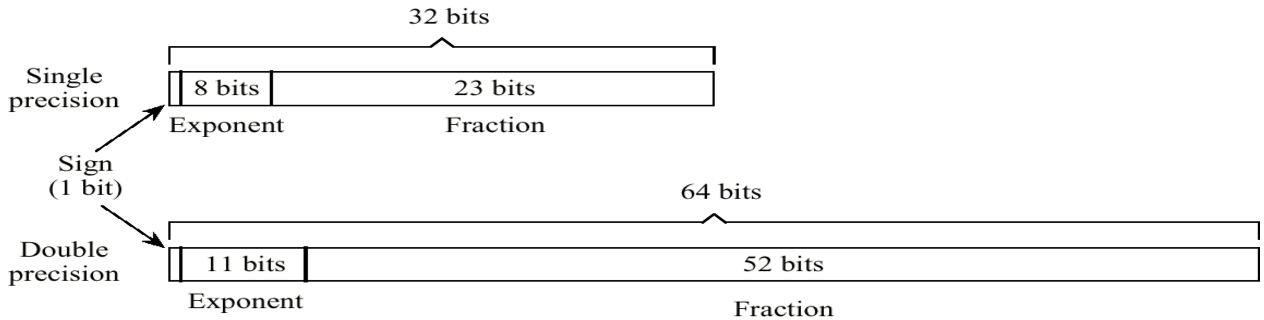
Question

Encoding the value -5.75 by IEEE 754 Standard



Question

Encoding the value -5.75 by IEEE 754 Standard



- -5.75->(1)101.11
- Left rotate 2bits: 1.0111

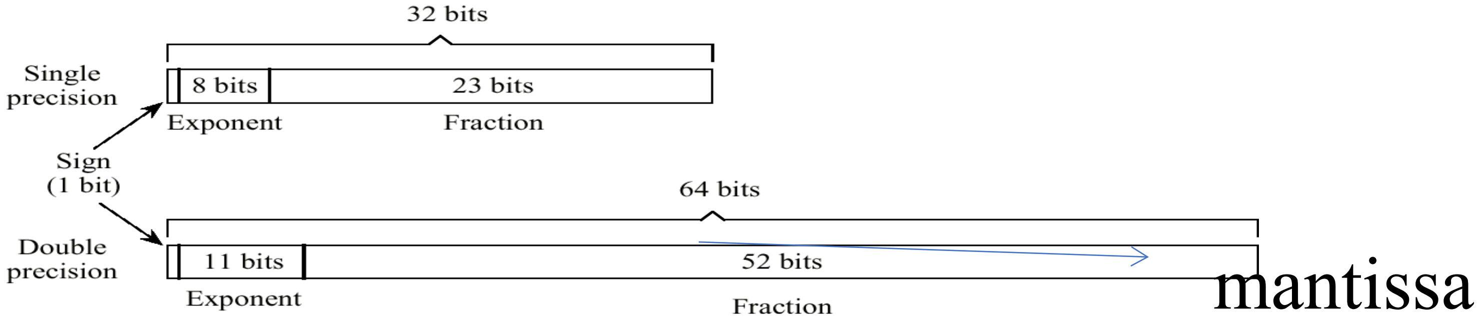
exponent=2->2+127->010+01111111=10000001

mantissa= 1.0111 -1= 0.0111

1 10000001 01110000000000000000000

$$(-1)^{sign\ bit} * (1 + mantissa) * 2^{(\mathrm{exponent}-127)}$$

Figure 1.27 Encoding the value 2 5/8->(10.101) ---- IEEE 754标准



178.125->10110010.001

Left rotate 7bits: 1.0110010001

exponent=7->7+127->111+01111111=10000110

mantissa= 1.0110010001 -1= 0.0110010001

0 10000110 01100100010000000000000

$$(-1)^{sign\ bit} * (1 + mantissa) * 2^{(\text{exponent}-127)}$$